DJANGO E-COMMERCE WEBSITE

# Nook & Nest

## Product Catalog, Cart, Orders & Inventory

A complete Django store with customer accounts and admin management.

Project documentation · September 2026

# 1. Overview

Nook & Nest is a full e-commerce web application for a curated home-goods store. It lets customers browse products, search and filter the catalog, add items to a session cart, sign in, check out, and view past orders. Administrators manage products, pricing, stock, categories, and order status using Django admin.

# 2. Feature mapping

| Requirement | Delivered capability |
| --- | --- |
| Product catalog | Category filters, search, product details, availability. |
| Cart | Session cart, quantities, removal, and calculated totals. |
| Orders | Authenticated checkout, line items, reference, and order history. |
| Authentication | Django registration, sign-in, sign-out, order ownership. |
| Payments | Demo Card workflow with a transaction reference. |
| Admin dashboard | Django admin for products, inventory, orders, and categories. |
| Inventory | Stock display, out-of-stock state, transaction-safe decrement. |

# 3. Architecture

> **Customer interface → Django views and forms → Models / transactions → SQLite database**
> **Admin interface → Product, stock, and order management**

## 4. Payment approach

> **Demo payment only.** The project never receives card data or processes real money. Checkout generates a `DEMO-...` reference so the order flow can be demonstrated safely.

For production, the next steps are: create a Stripe PaymentIntent or Razorpay Order on the server; use the provider's hosted checkout; verify a signed webhook; mark the order paid only after verification; and store all provider secrets in environment variables.

## 5. Test evidence

- Catalog output test.
- Cart insertion test.
- Authenticated checkout test that confirms inventory drops after the order.

Run with `.venv/bin/python manage.py test`.

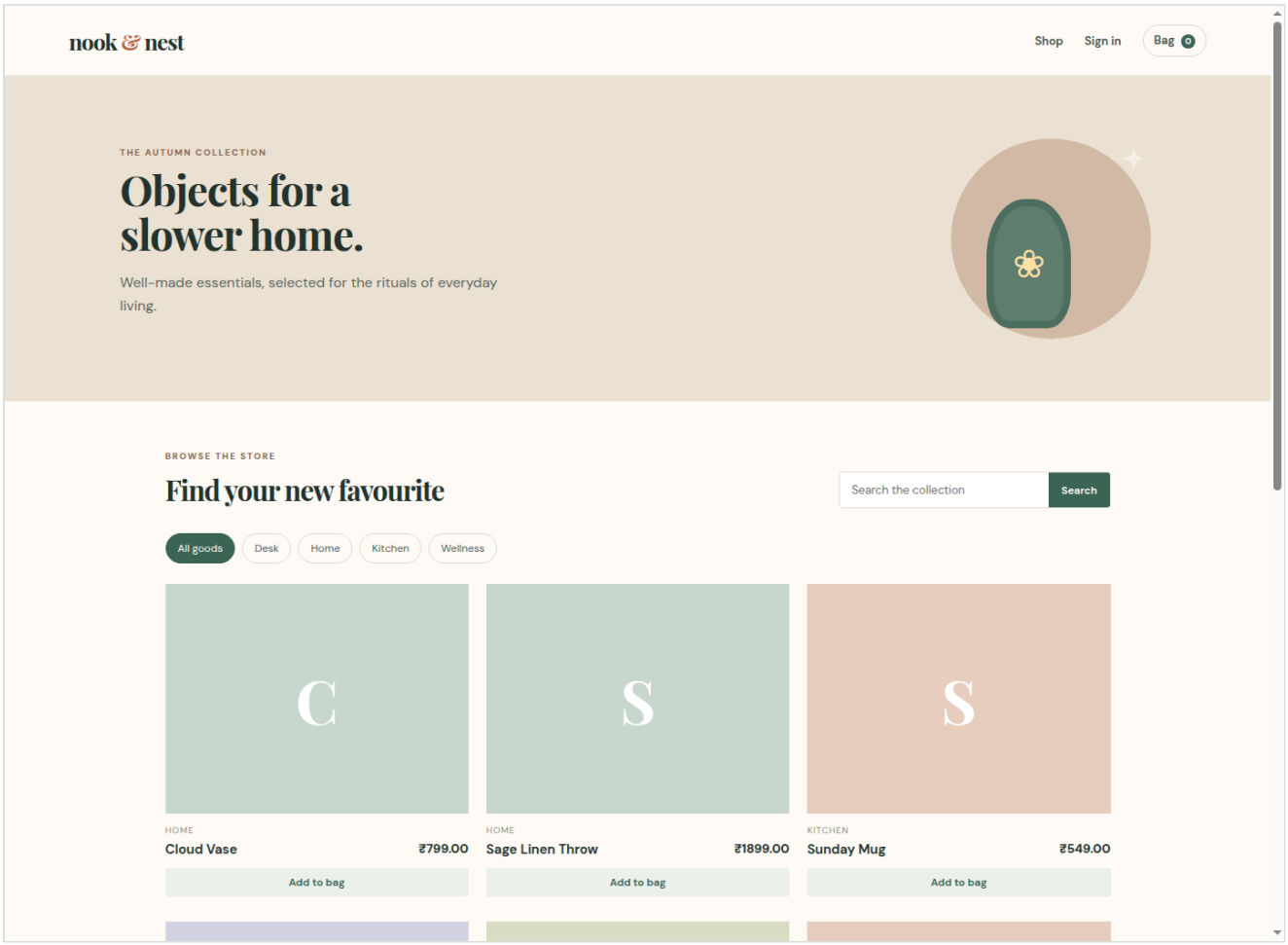
## 6. Run instructions

1. `.venv/bin/pip install -r requirements.txt`
2. `.venv/bin/python manage.py migrate`
3. `.venv/bin/python manage.py seed_store`
4. `.venv/bin/python manage.py runserver`

Create an administrator with `.venv/bin/python manage.py createsuperuser`, then open `/admin/`.

# 7. Application screens

## Catalog




## Cart

YOUR SELECTION

# Your bag




HOME

**Cloud Vase**

₹799.00

Quantity 1 Update

Set to 0 to remove

₹799.00

## Order summary

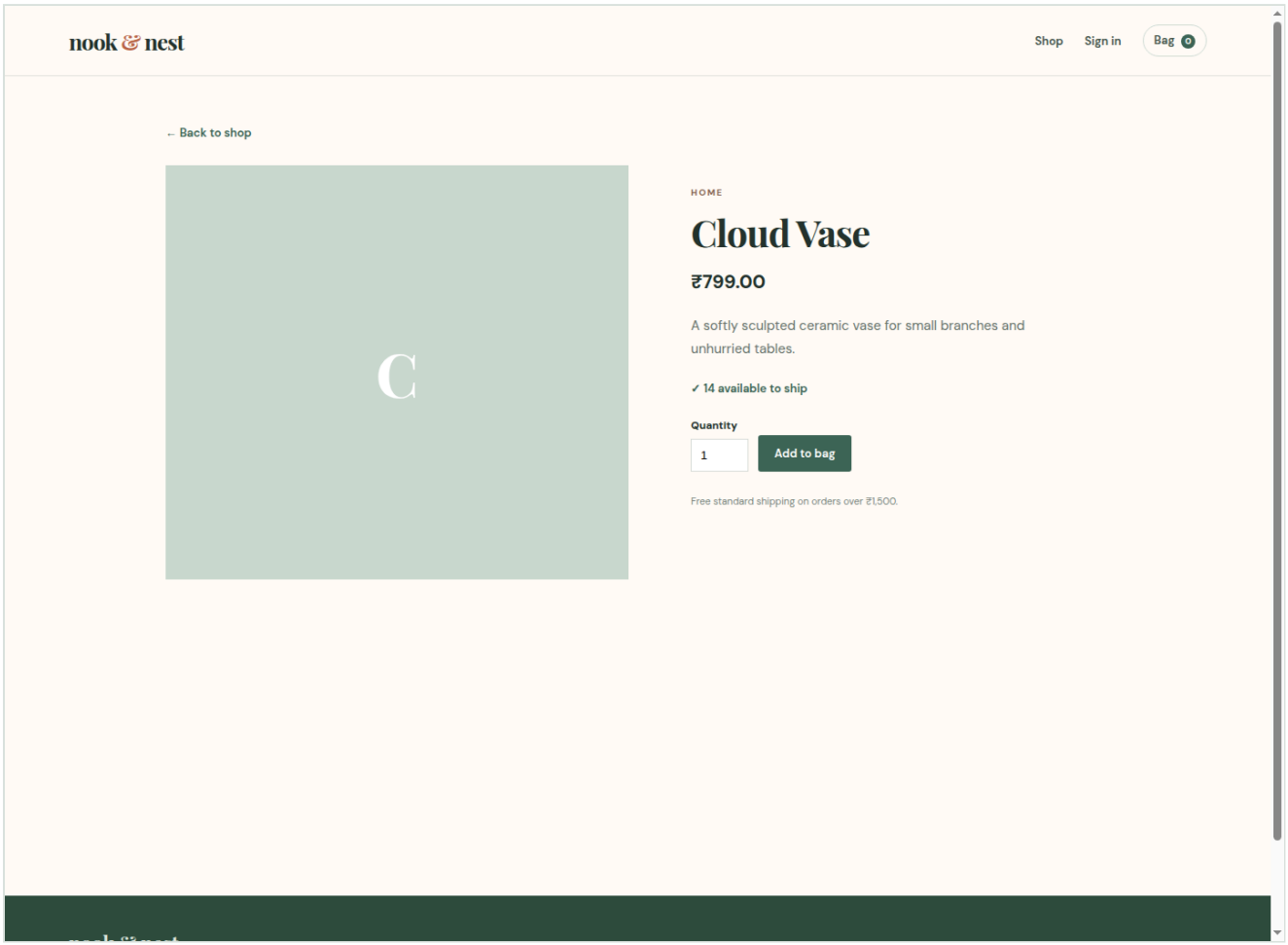
| | |
|---|---|
| Subtotal | ₹799.00 |
| Shipping | Free |
| Total | ₹799.00 |

Secure checkout

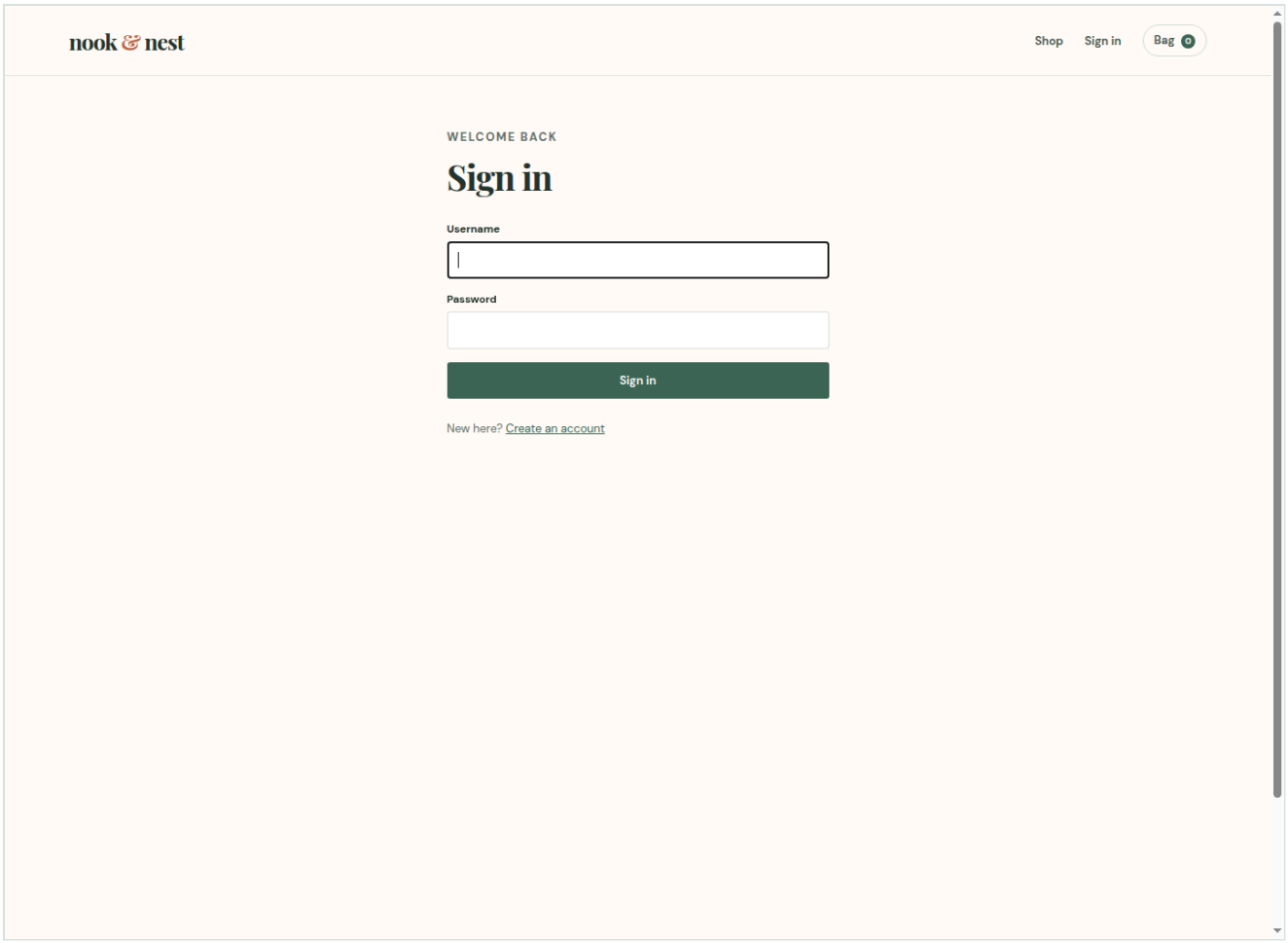
Demo checkout — no real payment is taken.

# 8. Application screens

## Product detail and inventory




## User authentication




# 9. Future scope

Future features include verified Stripe/Razorpay payments, coupon codes, customer reviews, shipment tracking, staff roles, sales analytics, and REST API access.